Incremental Source-Library Learning in CeTTa

A reproducible, frozen-profile PLN proof of concept

Mettapedia Working Notes

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Abstract

A native CeTTa learner discovers two reusable computations in an unchanged PLN truth-formula library. Successive retained programs contain 2,207, 2,186 and 2,180 structural nodes, counting helper definitions and calls. The experiment checks original-source reconstruction, finite public-call observations and stopped/resumed agreement. Its decisive language extension exposes one existing authored call, `/safe`, while leaving its definition unchanged. The extension is isolated in a retained, checksummed profile rather than silently installed into the default codec. This note records the algorithmic connection to incremental compression, the precise experiment, two semantic counterexamples, and interfaces for later integration. It closes a bounded experimental tranche, not a universal correctness or optimality claim.

1 What is being learned

The input is `lib/petta/lib_pln.metta` in the CeTTa NIK source tree. The task is *source-level library learning*: find reusable parameterized computations in existing truth-function implementations. It is not induction of new PLN inference rules, proof compression relative to a proof calculus, or fitting a numerical model to observations.

The native loop is

$$P_0 \longrightarrow (L_1, R_1) \longrightarrow (L_2, R_2), \qquad \text{expand}(L_i, R_i) = P_0.$$

Here P_0 is the admitted syntax document, L_i contains learned definitions, and R_i is the rewritten source containing calls and arguments. Each step starts a new search on the preceding stored program. The second helper need not call the first: successive independent abstractions are already incremental; hierarchical helper composition is a separate possible result.

Franz, Antonenko and Soletskyi describe compression by feature maps and descriptive maps, iterated over residual descriptions [1]. In this implementation anti-unification proposes features; matching and nonoverlapping replacement supply a restricted descriptive map; syntactic helper expansion reconstructs the observation. The practical search neither finds nor assumes a globally shortest feature. Independence, orthogonality and near-optimality conclusions requiring the paper's stronger hypotheses do not follow from the checks here.

There is also a useful distinction from residual-only presentations: later searches may inspect earlier helper definitions as part of the whole stored program. Thus the implementation permits joint library/residual optimization, not only a frozen feature sequence with residual-only updates.

2 Representation, observation and cost

The source representation retains expression structure, typed literals, form-local variable relationships, quotation, rule order and duplicate forms. Source reconstruction compares the normalized source emitted by the native reader/writer; it does not preserve original comments or layout byte for byte. The accumulated library expands newest-first to that original observation.

The selected objective is structural program size:

$$C(a) = 1 \quad \text{for an atom,} \qquad C((t_1 \dots t_n)) = 1 + \sum_{j=1}^{n} C(t_j).$$

An empty expression therefore costs one. The measured document includes every stored helper equation, parameter occurrence, call and delayed argument. Earlier helper definitions remain charged. Literal spelling, source layout and variable-inventory/codec metadata are outside this objective. In particular, long helper names do not receive a larger structural charge.

This is *not* a demonstrated complete binary MDL code for arbitrary literal payloads. Serialized archive lengths are reported separately. The profile and runtime retained for this experiment are supplied out of band and are not included in those archive lengths; the capsule's total size must not be mistaken for the table's encoded-program size.

Every accepted candidate must reconstruct the original observation and strictly lower the selected cost. Strict natural-valued descent bounds accepted updates but does not guarantee discovery, bounded checking time, semantic equivalence under all observations, or optimality of the selected feature.

3 The two learned computations

The first abstraction occurs four times:

```
(= (ic-learned:a $p0 $p1 $p2)
   (/ (* (+ 1.0 (/ $p0 $p1)) $p2)
      (+ 1.0 $p2)))
```

It factors the shared similarity-to-inheritance strength computation used inside transitive similarity. The second occurs in both `Truth_Deduction` and `simpleDeductionStrength`:

```
(= (ic-learned:b $p0 $p1 $p2 $p3 $p4 $p5)
   (if (< $p0 $p1)
       $p2
       (+ (* $p3 $p4)
          (/safe (* (- $p5 $p3)
                    (- $p2 (* $p1 $p4)))
                 (- $p5 $p1)))))
```

The arguments represent the authored threshold, intermediate and consequent strengths, the two conditional strengths, and the unit literal. Differences between `0.9999` and `0.99`, and between `1` and `1.0`, remain explicit arguments. No algebraic rewriting or literal normalization is used to force a match.

Stored program	Structural nodes	Archive bytes	Helpers retained
Original	2,207	59,537	0
After first feature	2,186	59,092	1
After second feature	2,180	59,059	2

The total reduction is 27 nodes (about 1.22%). The second reduction is small but fully charged: two 32-node subtrees become two 8-node calls and a 42-node definition,

$$2 \cdot 32 - (2 \cdot 8 + 42) = 6.$$

The significance is an inspectable second feature on real unchanged source, not a large compression ratio or an execution speedup.

4 Why the authored-call boundary mattered

The stock source profile exposes a fixed inventory of standard operators. Unknown calls become opaque captured expressions. Consequently the common structure inside `/safe` was not available to the template language; the associated capture/thunk overhead made the relevant candidate unprofitable.

The returned working review reports a complete census of the stock finite same-head pair language: 385 pairs and 41 distinct initial candidates; after the first feature, 229 pairs and 38 distinct candidates, none improving. The latter region's best candidate grows by five nodes. This is a reported finite-language census, not incompressibility under every possible grammar. Its Python/native diagnostic cross-check is external-review evidence; the local tranche qualification instead directly replays the positive experiment and the native controls below.

The experimental profile separates reserved primitive names from admitted authored calls and exposes `/safe` at arity two. It checks the exact source definition before learning:

```
(= (/safe $A $B)
   (if (> $B 0.0) (/ $A $B) (empty)))
```

Both zero and negative denominators produce no result. The extension must agree across traversal, projection, helper construction, argument recovery and decoder substitution. Adding a name only to the generator would not establish that agreement. Treating an authored call as a reserved intrinsic would instead reject its legitimate definition.

The fixed run selects the first four ranked seed pairs at stage one and the first pair at stage two, from expanded regions of 965 and 746 pairs. Templates and occurrence sites are generated, not supplied as witnesses. These prefixes were chosen during diagnosis; they are not an unbiased discovery benchmark or an exhaustive search of the expanded grammar. Multiple better incumbents can be considered within one stage: the recorded first stage has two such updates but retains one helper, not two independently installed features.

5 Qualification and counterexamples

The native shell-driven gate checks:

1. the original input hash, two retained helpers and their four/two uses;
2. native reconstruction to the original after each stage, with a separate AWK count agreeing with all three structural sizes;
3. 33 public-call observations on the stock runtime, covering threshold boundaries, empty and duplicate choice, entry effects and nonpositive safe division, plus a separate arithmetic-fault comparison;
4. stopped/resumed search at both stages yielding byte-identical artifacts;

5. 17 cost controls, the controlled two-step rule-head suite, four exact-call profile controls, malformed-archive refusal, and fresh-process capsule replay.

This command uses native CeTTa, shell tools and AWK, not a Python reconstruction checker. The observation panel is finite and does not prove universal execution equivalence.

Ambient type counterexample. An ambient result-type annotation can expose source syntax as a returned value:

```
(: audit-f (-> Number Atom))
```

Factoring three repeated arithmetic rules reconstructs exactly and reduces 75 nodes to 49, yet calling `(audit-f 3)` returns the arithmetic expression in the original and a helper-call expression in the rewrite. Source-local annotation checks alone do not establish that the target is untyped in the active environment.

Reflection counterexample. An untouched observer using `match &self` to inspect the body of a rewritten rule can distinguish it. The local negative control reduces 91 nodes to 65 and reconstructs exactly, but the observer's body-size result changes from 4 to 2. Avoiding transformations *inside* a reflective function does not prevent it observing transformations elsewhere.

These are reproduced limitations, not acceptable optimizations under arbitrary observations. The advertised POC contract is exact admitted-source reconstruction plus the specified ordinary public-call observations in fresh, fixed environments. General ambient target-type and helper-name freshness checks remain integration work. Reflective equivalence requires a separately defined reflection contract or decoding before the reflected execution.

6 Preserved package and extension interfaces

The ordinary libraries and C dispatch are unchanged. An opt-in experiment copies the libraries, applies the two-file profile patch there, and retains the matching decoder, runtime and artifacts in a fresh capsule. The input is pinned by its full SHA-256. `PROFILE.txt`, `PROFILE.sha256` and `ARTIFACTS.sha256` travel with the artifacts; replay verifies them before decoding in a fresh process. These checks provide an integrity/reproducibility envelope, not an authenticated trust boundary.

In particular, a raw experimental `LC2/LM1` archive does not yet carry a general versioned authored-call profile. Its matching capsule is required. The emitted ordinary MeTTa definitions themselves run on the stock runtime. This difference between executable program export and archive decoding is an explicit boundary of the paused tranche.

```
make BUILD=core ENABLE_LIB_PROLOG=0 CETTA_TEST_ISOLATED=1 demo-ic-pln-mvp
bash experiments/ic_pln_mvp/replay.sh path/to/capsule
bash experiments/ic_pln_mvp/check_boundaries.sh path/to/cetta
```

The last command succeeds when the two negative counterexamples reproduce; it is not a general semantic-correctness gate.

The principal extension points are:

Persisted profile. Keep the profile explicit throughout preparation, decoding and resumption. It records reserved primitives, authored signatures and definitions, argument demand, the environment and the observation. Changed dependencies must invalidate or reprepare

the task. Ambient target-type and namespace checks must be fixed before broadening automatic installation.

Exact planning before publication. Select actual disjoint replacement sites, then compute

$$C_{\text{next}} = C_{\text{current}} + C_{\text{definition}} + \sum_{s \in S} (C_{\text{call(s)}} - C_{\text{old(s)}}).$$

Disjointness makes this an exact structural delta, including actual thunks. Only improving plans need whole-document packing; the final checker remains authoritative. The current materialization interface receives the stage input, not the incumbent: incumbent-based pruning needs its validated cost explicitly.

Prepared state and deduplication. Reuse catalogs within a stage. Any candidate key must retain sorts/roles, shared-hole relationships, readiness and the complete frozen context. Preserve first representatives and search order in differential replay. A similar printed body is not a sufficient key.

These are future integration tasks, not additional acceptance gates for the retained experiment. Broader numerical learning, proof-trace compression, ILP metarule discovery and hierarchical helper invention remain separate tasks.

7 Theory connection and evidence provenance

Mettapedia contains shared source-frame, template, costed-artifact, layered source and search developments under `Mettapedia/Algorithms/Compression/`, and paper-facing definitions under `Mettapedia/Computability/KolmogorovComplexity/`. The concrete `SourceAdmissionCounterexample.lean` models the older plain-port restriction and proves the empty-list predicate's break-even calculation. Those model results do not prove the current C runtime equivalent to Lean reduction or establish all source transformations semantically sound. No new Lean theorem is asserted by this experimental handoff.

The working diagnostic originated in the returned 30 October 2026 review; local packaging adds the native observation-boundary runner and replay envelope. The base CeTTa revision was `45d3b058`, with the retained source-runner changes. The input hash is:

```
6b980321bbe9b49e5b12e2fcee0479ab1d5c7550c2104a67b0722a5b473da4ee
```

The returned review archive hash is:

```
b705188c3807c6f8fdaee8ac32684a9b3981bc0184e7327c88e393c28545ec2c
```

Executable source, commands and integration notes are owned by `experiments/ic_pln_mvp/`. Each run retains its own binary/library hashes and check outputs. Historical timings from different machines, binaries and search-region sizes are not a controlled speed comparison and are not used to claim a speedup here.

The defensible conclusion is a small real-program incremental abstraction result with exact source reconstruction and an explicit execution profile. It is sufficient to pause and preserve this tranche while leaving its broader semantic and production extensions clearly identified.

References

[1] Arthur Franz, Oleksandr Antonenko and Roman Soletskyi. *A theory of incremental compression*. Information Sciences 547 (2021), 28–48. https://arxiv.org/abs/1908.03781v2.